

RESEARCH ARTICLE

Acute and chronic effects of hot water immersion on inflammation and metabolism in sedentary, overweight adults

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Submitted 10 May 2018; accepted in final form 14 October 2018

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Acute and chronic effects of hot water immersion on inflammation and metabolism in sedentary, overweight adults. *J Appl Physiol* 125: 2008–2018, 2018. First published October 18, 2018; doi: 10.1152/jappphysiol.00407.2018.—Regular exercise-induced acute inflammatory responses are suggested to improve the inflammatory profile and insulin sensitivity. As body temperature elevations partly mediate this response, passive heating might be a viable tool to improve the inflammatory profile. This study investigated the acute and chronic effects of hot water immersion on inflammatory and metabolic markers. Ten sedentary, overweight men [body mass index (BMI): 31.0 ± 4.2 kg/m², mean $\pm$ SD] were immersed in water set at 39°C for 1 h (HWI) or rested for 1 h at ambient temperature (AMB). Venous blood was obtained before the session, immediately postsession, and 2 h postsession for assessment of monocyte intracellular heat shock protein-72 (iHsp72) and plasma concentrations of extracellular Hsp72 (eHsp72), interleukin-6 (IL-6), fasting glucose, insulin, and nitrite. Thereafter, participants underwent a 2-wk intervention period, consisting of 10 hot water immersion sessions (INT). Eight BMI-matched participants (BMI: 30.0 ± 2.5 kg/m²) were included as control (CON). Plasma IL-6 and nitrite concentrations were higher immediately following HWI compared with AMB (IL-6 $P < 0.001$, HWI: 1.37 ± 0.94 to 2.51 ± 1.49 pg/ml; nitrite $P = 0.04$, HWI: 271 ± 52 to 391 ± 72 nM), whereas iHsp72 expression was unchanged ($P = 0.57$). In contrast to resting iHsp72 expression ($P = 0.59$), fasting glucose ($P = 0.04$; INT: 4.44 ± 0.93 to 3.98 ± 0.98 mmol/l), insulin ($P = 0.04$; INT: 68.1 ± 44.6 to 55.0 ± 29.9 pmol/l), and eHsp72 ($P = 0.03$; INT: $17 \pm 41\%$ reduction) concentrations were lowered after INT compared with CON. HWI induced an acute inflammatory response and increased nitric oxide bioavailability. The reductions in fasting glucose and insulin concentrations following the chronic intervention suggest that hot water immersion may serve as a tool to improve glucose metabolism.

NEW & NOTEWORTHY A single hot water immersion (HWI) session induces an acute increase in plasma interleukin-6 and nitrite concentrations but does not acutely elevate heat shock protein-72 expression in monocytes [intracellular Hsp72 (iHsp72)]. A chronic HWI intervention reduces fasting glucose and insulin concentrations in the absence of changes in resting iHsp72. Therefore, HWI shows potential as a strategy to combat chronic low-grade inflammation and improve glucose metabolism in individuals without the physical capacity to do so using exercise.

chronic low-grade inflammation; glucose metabolism; heat shock protein; interleukin-6; passive heating

INTRODUCTION

Passive heating interventions have been linked to several positive health outcomes, such as improved vascular function (4) and mental health (11), weight loss (33), and enhanced insulin sensitivity (42). Although observations of a lowering in fasting glycosylated hemoglobin and blood glucose concentrations following hot water immersion (HWI) in individuals with type 2 diabetes support the notion of improved insulin sensitivity following HWI (33), the mechanisms that underlie this beneficial effect are currently unclear. Chronic low-grade inflammation has been implicated in the etiology of insulin resistance (9), as evidenced by the positive association between proinflammatory proteins and insulin resistance (9, 39), while the body of evidence for a causal relationship of these proteins with insulin resistance is growing (35). Moreover, it is well documented that exercise training can counteract chronic low-grade inflammation (57) and improve insulin sensitivity (29). However, since it is not feasible for all populations to adhere to the recommended exercise guidelines because of a low physical capacity or health conditions that hinder exercise participation, the development of alternative strategies that can reduce chronic low-grade inflammation in populations without the capacity to engage in sufficient volumes of exercise is warranted to mitigate risk factors for insulin resistance and noncommunicable diseases.

The acute inflammatory response provoked by a physical stressor, such as exercise, can induce a subsequent protracted anti-inflammatory response. For instance, elevations in circulating interleukin-6 (IL-6) concentrations immediately following exercise activate the release of anti-inflammatory cytokines such as IL-1 receptor antagonist (IL-1ra) and IL-10, typically 1–4 h following the exercise bout (57). In addition, recent studies have identified an enhanced acute inflammatory response following exercise when body temperature is augmented (43). Increasing body temperature therefore likely serves as an independent stressor able to induce the acute inflammatory responses needed to reduce chronic low-grade inflammation in the long term. This is supported by Welch et al. (66), who showed that passive heating for 1 h at 42.4°C can activate heat shock factor 1, which in turn upregulates the

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Table 1. Participant characteristics at baseline

Parameter	INT	CON	P (ES)
<i>n</i>	10	8	
Age, yr	33.2 ± 10.1	32.0 ± 7.5	0.39 (0.14)
Height, cm	173.2 ± 6.9	176.9 ± 4.2	0.17 (0.67)
Body mass, kg	92.1 ± 9.2	93.9 ± 9.4	0.36 (0.21)
BMI, kg/m ²	31.0 ± 4.2	30.0 ± 2.5	0.37 (0.30)
Fat percentage, %	25.1 ± 3.5	25.7 ± 2.3	0.23 (0.18)
Waist circumference, cm	96.9 ± 4.7	100.1 ± 8.4	0.14 (0.52)
Hip circumference, cm	103.9 ± 4.1	109.4 ± 5.1	0.08 (1.28)
Structured exercise, min/wk	38 ± 54	29 ± 44	0.49 (0.19)

Values are means ± SD or *P* values with Cohen's *d* effect size (ES) in parentheses; *n* = no. of participants. BMI, body mass index; CON, control group; INT, intervention group.

production of IL-6 and intracellular heat shock protein-72 (iHsp72) in mouse skeletal muscle.

In humans, 1–2 h of HWI, at a temperature 2–3°C higher than resting core temperature, has been reported to acutely elevate IL-6, IL-1ra (45), extracellular Hsp72 (eHsp72; 16), and monocyte iHsp72 (54). Elevations in iHsp72 can block the inflammatory actions of c-Jun NH₂-terminal kinase (JNK) and nuclear factor-κB (NF-κB), resulting in enhanced insulin sensitivity (31). In contrast to the beneficial functions of iHsp72, Hsp72 found in plasma (i.e., eHsp72) can activate circulating monocytes, resulting in an increase in proinflammatory cytokine release (1). Although the transient increase in eHsp72 following an acute bout of exercise is suggested to be part of the beneficial inflammatory response to exercise (67), a reduction in resting eHsp72 is suggestive of an improved inflammatory profile and may improve insulin sensitivity (41).

In addition to modulating inflammation, an increase in body temperature has been linked to increased nitric oxide (NO) production through enhanced NO synthase (NOS; 4, 36), possibly mediated by an increased expression of Hsp90 (70). It is well documented that NO impacts a myriad of biological processes, including tissue glucose uptake (19, 20, 58, 60). Therefore, an increase in NO synthesis following HWI might contribute to changes in insulin sensitivity resulting from this intervention. Moreover, an acute increase in NO bioavailability exerts an anti-inflammatory effect on human leukocytes (58) and increases the iHsp72 expression in peripheral mononuclear blood cells (63), indicating cross talk between NO and the immune system. However, the extent to which acute and chronic HWI influences NO synthesis and its role in chronic low-grade inflammation and insulin sensitivity are presently unclear.

Although there is now evidence for the potential of HWI to induce an acute inflammatory response (16, 45, 54), chronic intervention studies in humans are scarce. Notwithstanding, the reduction in fasting blood glucose concentrations in patients with diabetes (33) and the reduction in resting plasma IL-6 concentrations in patients with chronic heart failure (55) are promising initial results. These studies, however, focused on clinical populations, did not address the mechanistic link between inflammatory and metabolic markers, and provided little detail on the acute (thermo)physiological responses to HWI. For instance, whereas animal studies have provided compelling evidence for the potential of HWI to chronically elevate basal iHsp72 levels (6, 26, 61), it is not known whether this holds true in humans. The smaller acute core temperature increases reported in human compared with animal studies might make HWI less effective as a strategy to elevate resting iHsp72 levels in humans (27).

Therefore, the present study investigated the acute inflammatory response to a single HWI session as well as the potential of a chronic HWI intervention to improve the inflammatory and metabolic profile at rest. It is hypothesized that an HWI session induces acute increases in plasma IL-6 concentrations, NO bioavailability, and iHsp72 expression in monocytes. Chronically, the 2-wk HWI intervention is hypothesized to increase resting levels of iHsp72, while reducing IL-6 and eHsp72 concentrations. Finally, in line with Hooper (33), the intervention period is expected to result in reductions in fasting glucose and insulin concentrations.

METHODS

Participants. Participants were sedentary (<2 h exercise/wk), overweight (body mass index >27 kg/m²), otherwise healthy men (Table 1). Exclusion criteria were the usage of anti-inflammatory medication and contraindications to engage in HWI. The latter was assessed with a medical health questionnaire according to the American College for Sport and Exercise Medicine guidelines for exercise testing and prescription (32). Engagement in structured exercise was reported before and following the chronic intervention period, using the International Physical Activity Questionnaire (8). Participants gave written informed consent after being instructed about the procedures of the study, which were approved by the Local Ethical Committee of Loughborough University, in accordance with the Declaration of Helsinki.

Procedures. An outline of the procedures for the intervention group is given in Fig. 1. Participants visited the laboratory for a HWI (HWI_{pre}) and control trial (AMB) in a counterbalanced order, with a minimum of 72 h between the visits. Participants refrained from exercise, alcohol, and caffeine and standardized their diet using a food

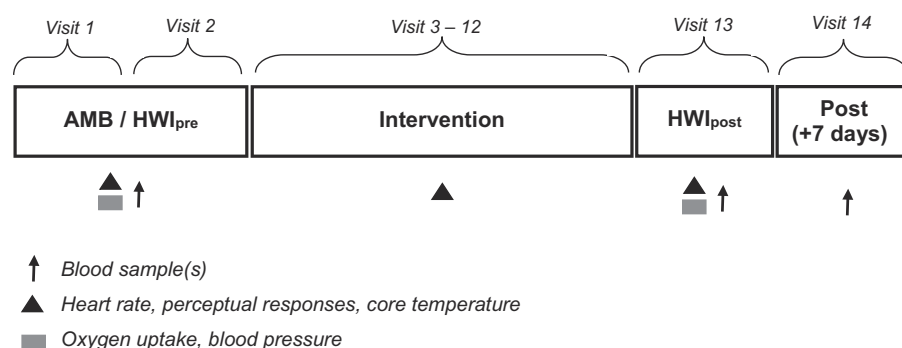


Fig. 1. Outline of the study procedures for the intervention group. An acute hot water immersion (HWI_{pre}) and control trial (AMB) were followed by 10 HWI sessions within 2 wk. A second acute HWI trial (HWI_{post}) was conducted 3 days after completion of the intervention period, and a resting blood sample was taken 7 days following HWI_{post} (Post). For the control group, a resting blood sample was taken at the time points corresponding to visits 1 and 13 of the intervention group.

diary in the 24 h before the visits. All visits started between 8 and 10 AM, with the starting time consistently applied for each individual to account for a possible circadian rhythm in any of the outcome measures. After an overnight fast, nude body mass, height, and hip and waist circumferences were measured, and skinfold thickness was assessed at four sites (biceps, triceps, subscapular, and suprailiac; 14) for the estimation of body fat percentage.

Thereafter, participants underwent 15 min of seated rest in an environmental chamber (27°C, 40% humidity) for baseline measurements (21). Following the "Pre" blood sample, participants entered the water tank for the HWI_{pre} or remained seated for another hour in the same conditions for AMB. This control condition (instead of immersion in thermoneutral water) was chosen because this study was designed to evaluate the effects of HWI as a stand-alone health intervention rather than to investigate the effects of an increase in body temperature per se. Evidence suggests that the effects of hydrostatic pressure on inflammatory markers are negligible (43).

During HWI_{pre}, participants were immersed up to the neck for 1 h in water set at 39°C. Participants sat in an upright position and were allowed to drink water ad libitum. During both HWI_{pre} and AMB, measurements were taken every 15 min. Blood pressure (Microlife BP3AC1-1; Microlife, Cambridge, United Kingdom) was measured in duplicate at the level of the heart, while thermal sensation, thermal comfort (21), and basic affect using the Feeling Scale (68) were reported. Expired air was collected for 3 min into Douglas bags for the determination of oxygen uptake using a Servomex 1440 gas analyzer (Servomex, Crowborough, United Kingdom). Tympanic temperature was measured with a tympanic temperature probe (Squirrel; Grant Instruments, Shepreth, United Kingdom), using cotton wool to cover the external canal of the ear. Rectal temperature was recorded every 5 min throughout the trials, using a rectal probe (YSI 400 series; YSI, Yellow Springs, OH) that was inserted 10 cm beyond the anal sphincter. Heart rate (HR, Polar RS400; Polar, Kempele, Finland) was continuously measured throughout.

Immediately on completion of the session, a "Post" blood sample was taken, and participants rested seated in the environmental chamber for 30 min. Thereafter, nude body weight was measured, and a breakfast snack was provided (Sainsbury's breakfast biscuits; 212 kcal, 5.8 g fat, 34.3 g carbohydrates, and 4.0 g protein). The change in nude body weight and water consumed was used to estimate sweat loss. Participants were then allowed to rest and perform light work such as reading. Two hours after completion of the session, the "Post+2h" blood sample was taken following 15 min of seated rest.

Following the first 2 visits, participants enrolled in an intervention period consisting of 10 HWI sessions, all executed within 14 days. The first five sessions of this period lasted 45 min, whereas the last five lasted 60 min. As pilot work suggested that the HWI sessions can be experienced as uncomfortable, this progression was chosen to avoid drop-out during the intervention period. In all sessions the temperature of the water was set at 39°C, and participants were immersed up to their neck. During the 10 sessions, HR, tympanic temperature, thermal sensation, thermal comfort, and basic affect were assessed every 15 min. Three days after completion of the last session of the intervention period, an acute trial (HWI_{post}) was conducted to study the effects of the intervention period on the acute inflammatory response to HWI. The procedures during this session were identical to HWI_{pre}. The Pre blood samples of the first session (either HWI_{pre} or AMB) and HWI_{post} were used to study the chronic effects of the intervention period. Eight individuals matched for body composition, age, and physical activity levels were included as control for the chronic arm of the study (CON). These participants visited the laboratory for two resting blood samples only, with the time between both samples held equal to the intervention group. In the intervention group, an additional resting blood sample was taken 1 wk following HWI_{post} ("Post+1wk") to investigate whether any adaptations detected following the intervention period would remain after 1 wk.

Biochemical analyses. Blood was collected in tripotassium EDTA (K₃EDTA; plasma markers) and sodium heparin (flow cytometry) Monovettes. The K₃EDTA tubes were spun down immediately for 5 min at 1,500 g and 4°C, and plasma was stored at -80°C until batch analysis. Flow cytometry was used to assess changes in iHsp72 in monocytes and the distribution in monocyte subsets. In addition, changes in the expression of iHsp72 in the respective monocyte subsets were assessed. Sixty microliters of whole blood were incubated together with 5 µl of peridinin chlorophyll protein-conjugated cluster of differentiation 14 (CD14) and 2.5 µl of phycoerythrin-conjugated CD16 antibodies in the dark at room temperature for 15 min. Thereafter, samples were lysed (750 µl, FACS lysing solution; BD Biosciences, San Diego, CA), washed (1.5 ml phosphate-buffered saline), and fixed using Leucoperm (60 µl, BD Biosciences). Following permeabilization (60 µl, Leucoperm; BD Biosciences), samples were incubated with 4 µl of FITC-conjugated Hsp70 antibody or isotype control for 30 min. Finally, samples were washed and resuspended in phosphate-buffered saline before running through the Flow Calibur (BD Biosciences). All antibodies except CD16 (BD Biosciences) were purchased from Miltenyi Biotec (Teterow, Germany). CellQuest software (BD Biosciences) was used for the analysis, collecting 100,000 events per sample. Compensation of the flow cytometer before the study was performed manually using a whole blood sample of a male volunteer not participating in the study. Monocytes were selected on the basis of positive CD14 expression, after which the percentage of monocyte subsets (CD14⁺⁺CD16⁻ classical monocytes, CD14⁺CD16⁺ intermediate monocytes, and CD14⁻CD16⁺⁺ nonclassical monocytes) was determined using the trapezoid method (71). The iHsp72 expression in monocytes was determined using the geometric mean fluorescence intensity (GMFI) following subtraction of the isotype control GMFI.

All glassware, utensils, and surfaces were rinsed with deionized water to remove residual NO intermediates before plasma nitrite concentration analysis. Plasma samples were introduced to a gas-tight purge vessel via 200-µl injections into the septum at the top of the vessel. The nitrite concentration of the undiluted (nondeproteinized) plasma was determined by its reduction to NO in the presence of glacial acetic acid and aqueous sodium iodide (4% wt/vol). The spectral emission of electronically excited nitrogen dioxide, derived from the reaction of NO with ozone, was detected by a thermoelectrically cooled, red-sensitive photomultiplier tube housed in a Sievers gas-phase chemiluminescence nitric oxide analyzer (Sievers NOA 280i; Analytix, Durham, United Kingdom). The nitrite concentration was determined by plotting signal (mV) area against a calibration plot of sodium nitrite standards. IL-6 (high-sensitivity; R&D Systems, Abingdon, United Kingdom), eHsp72 (AMP'D Hsp70 high-sensitivity; Enzo Life Sciences, Farmingdale, NY), and insulin (Mercodia, Uppsala, Sweden) were measured in plasma, in duplicate, using enzyme-linked immunosorbent assays (ELISA). For the determination of eHsp72 concentrations, plasma samples were diluted 1:4 before running the ELISA. The intra-assay coefficients of variation were 7.0, 6.2, and 2.5% for IL-6, eHsp72, and insulin, respectively. A Biosen C-Line (Biosen, Barleben, Germany) was used to determine blood glucose concentrations in whole blood (52). A whole blood count was obtained using a Yumizen H500 cell counter (Horiba Medical, Montpellier, France) for the determination of leukocyte subsets, hematocrit, and hemoglobin. The latter two were used to correct the Post and Post+2h plasma IL-6 and eHsp72 concentrations for changes in plasma volume (10).

Statistical analyses. All values are given as means ± SD. Normality of the data was checked using the Shapiro-Wilk test, and a log transformation was performed when nonnormality was detected. Log transformation was performed on the eHsp72 data. Analysis of variance with repeated measures where appropriate was used to detect differences in the acute responses between AMB and HWI_{pre} and between HWI_{pre} and HWI_{post} as well as the effects of the intervention period on baseline measures compared with CON. Because of a

Table 2. *Physiological responses to the hot water immersion and control trial*

Parameter	AMB		HWI _{pre}		HWI _{post}	
	Pre	60 min	Pre	60 min	Pre	60 min
T _{rec} , °C	36.9 ± 0.5	36.6 ± 0.5	37.1 ± 0.6	38.7 ± 0.4*	37.0 ± 0.3	38.5 ± 0.3*
T _{tymp} , °C	35.5 ± 0.4	35.2 ± 0.4	35.4 ± 0.7	37.5 ± 0.7*	35.2 ± 0.3	37.6 ± 0.3*
HR, beats/min	69 ± 17	64 ± 15	67 ± 14	105 ± 13*	68 ± 12	104 ± 10*
VO ₂ , l/min	0.20 ± 0.04	0.23 ± 0.04	0.21 ± 0.04	0.42 ± 0.10*	0.19 ± 0.05	0.37 ± 0.03*
SBP, mmHg	127 ± 10	123 ± 12	126 ± 13	138 ± 15*	118 ± 15†	126 ± 13†
DBP, mmHg	86 ± 9	85 ± 10	83 ± 9	78 ± 9	79 ± 11†	71 ± 12*†
Basic affect (−5 to +5)	0.9 ± 1.4	0.7 ± 1.3	1.3 ± 2.0	−1.1 ± 2.2*	1.1 ± 1.9	−1.2 ± 1.9*
TS (1 to 9)	4.9 ± 0.6	4.8 ± 0.8	5.1 ± 0.9	7.4 ± 1.0*	4.7 ± 0.5	6.7 ± 0.8*†
TC (−5 to +5)	0.0 ± 0.0	0.0 ± 0.5	0.2 ± 0.4	2.2 ± 1.0*	−0.1 ± 0.3	2.3 ± 1.3*
Sweat loss, liters	N/A	0.17 ± 0.19	N/A	1.12 ± 0.56*	N/A	1.65 ± 0.57*†
PV change, %	N/A	98 ± 3	N/A	95 ± 8	N/A	95 ± 4

Values are means ± SD. AMB, control trial; DBP, diastolic blood pressure; HR, heart rate; HWI_{post}, hot water immersion (HWI) session following HWI intervention period; HWI_{pre}, HWI session before HWI intervention period; N/A, not applicable; Pre, before HWI or AMB; PV, plasma volume; SBP, systolic blood pressure; TC, thermal comfort (higher TC scores reflect reduced feelings of thermal comfort); T_{rec}, rectal temperature; TS, thermal sensation; T_{tymp}, tympanic temperature; VO₂, oxygen uptake. *Significantly different from AMB; †significant difference between HWI_{pre} and HWI_{post}.

difference in baseline plasma nitrite concentrations between HWI_{pre} and AMB, a one-way analysis of covariance was employed to detect differences between HWI and AMB at Post and Post+2h using nitrite concentrations at Pre as a covariate. *R*, the fold change in the eHsp72-to-iHsp72 ratio, was determined for the acute as well as the chronic arm of the study (41). The homeostatic model assessment for insulin resistance (HOMA-IR) was determined using fasting glucose and insulin concentrations (47). For all analyses, a Bonferroni-corrected post hoc test was used for exploration of the differences at every time point when significance was detected. Effect sizes (ES; Cohen's *d*) and their 95% confidence intervals were calculated where appropriate, whereby an ES of 0.20, 0.50, or 0.80 refers to a small, moderate, or large effect, respectively (7). The ES for a time × group (T × G) or time × condition (T × C) interaction was calculated by comparing the Pre-Post change scores in each group or condition. Correlations were computed using Pearson's *r*. As the latter was an explorative analysis, the risk for a type II error was not deemed problematic, and no Bonferroni correction was applied (56). The 23rd version of the statistical package SPSS Statistics (SPSS, Chicago, IL) was used for all analyses, and statistical significance was set at *P* < 0.05.

RESULTS

Participants. Baseline characteristics of the participants in the intervention group (INT) and CON can be seen in Table 1. Apart from a trend toward a larger hip circumference in CON, there were no differences in anthropometrics and physical activity levels between the groups.

Acute responses to HWI. The physiological and perceptual responses during HWI_{pre}, HWI_{post}, and AMB are given in Table 2. During HWI_{pre}, rectal temperature increased from 37.1 ± 0.6 to 38.7 ± 0.4°C (Fig. 2). Following the intervention period, diastolic blood pressure was lower at the end of HWI_{post} compared with HWI_{pre} (*F*: 25.4, *P* = 0.001). Thermal sensation at the end of HWI_{post} was lower than at the end of HWI_{pre} (*F*: 14.3, *P* = 0.01), and sweat loss during HWI was increased from 1.1 ± 0.6 (HWI_{pre}) to 1.7 ± 0.6 liters (HWI_{post}) (*F*: 26.5, *P* = 0.001).

Plasma concentrations of IL-6 were higher compared with AMB immediately following HWI_{pre} [T × C; *F*: 14.5, *P* < 0.001, ES: 1.71 (1.31–2.07), mean (95% confidence interval)]. However, this was not accompanied by a rise in either eHsp72 (T × C; *F*: 1.9, *P* = 0.16) or iHsp72 in total monocytes (T × C;

F: 0.5, *P* = 0.57) directly Post or Post+2h HWI_{pre} (Fig. 3). The same was true for the expression of iHsp72 in classical monocytes (T × C; *F*: 1.7, *P* = 0.22), intermediate monocytes (T × C; *F*: 2.3, *P* = 0.19), and nonclassical monocytes (T × C; *F*: 1.5, *P* = 0.25). *R* did not differ between HWI_{pre} and AMB (T × C; Pre-Post *F*: 0.6, *P* = 0.48; Pre-Post+2h *F*: 0.1, *P* = 0.76).

The distribution of monocyte subsets changed immediately after HWI_{pre}, with an increase of the intermediate [T × C; *F*: 9.0, *P* = 0.004, ES: 1.39 (0.36–2.03)] and nonclassical monocytes [T × C; *F*: 11.8, *P* = 0.001, ES: 1.34 (0.32–1.24)]. The proportion of classical monocytes, however, was not reduced (T × C; *F*: 2.5, *P* = 0.10; Table 3). Lymphocyte numbers increased to a larger extent directly following HWI_{pre} compared with AMB [T × C; *F*: 11.0, *P* = 0.003, ES: 1.97 (0.84–2.94)]. There was no difference between HWI_{pre} and AMB in the acute elevation of total monocyte (T × C; *F*: 0.8, *P* = 0.56), leukocyte (T × C; *F*: 2.0, *P* = 0.16), or neutrophil numbers (T × C; *F*: 2.7, *P* = 0.08). The increase in plasma nitrite concentration directly following HWI_{pre} was larger compared with AMB [*F*: 11.2, *P* = 0.04, ES: 1.82 (0.71–2.77); Fig. 3].

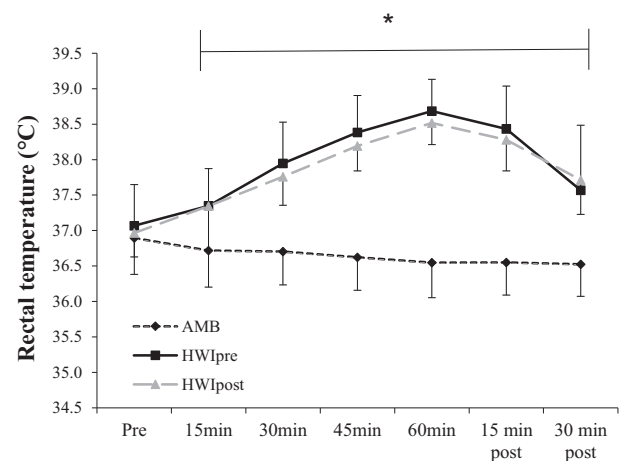
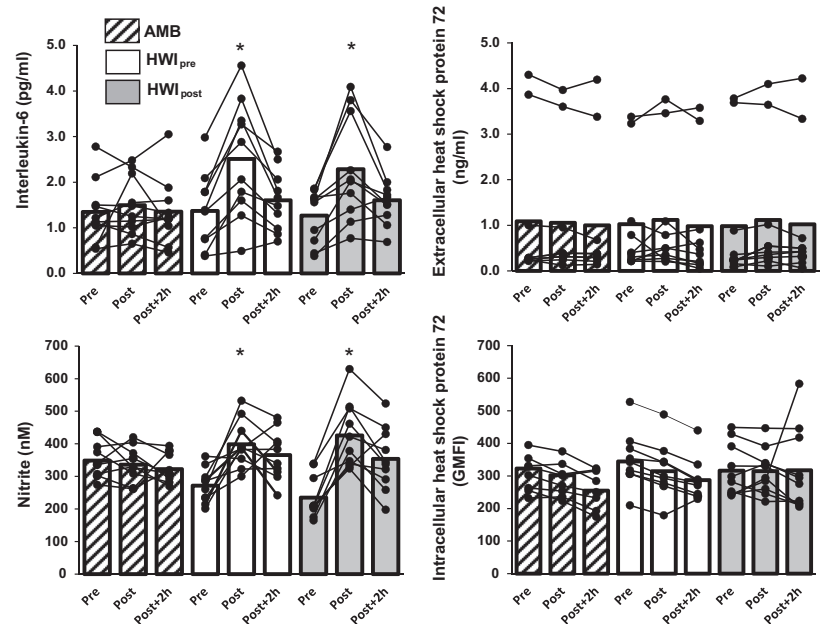


Fig. 2. Rectal temperature during and following control trial (AMB), hot water immersion (HWI) session before HWI intervention period (HWI_{pre}), and HWI session following HWI intervention period (HWI_{post}) (*n* = 10). *Significantly different from AMB.

Fig. 3. Acute changes in plasma IL-6, extracellular heat shock protein-72 (eHsp72), intracellular Hsp72 (iHsp72), and nitrite concentrations following control trial (AMB), hot water immersion (HWI) session before HWI intervention period (HWI_{pre}), and HWI session following HWI intervention period (HWI_{post}). GMFI, geometric mean fluorescence intensity. Black lines represent individual data points, whereas bars represent the group mean ($n = 10$). *Significant time $\times$ trial interaction compared with AMB.



The IL-6, eHsp72, and iHsp72 response did not differ following HWI_{post} compared with HWI_{pre} ($T \times C$; IL-6 F : 0.3, $P = 0.80$, eHsp72 F : 0.9, $P = 0.45$, iHsp72 F : 0.1, $P = 0.71$). The same was true for Hsp72 expression in classical ($T \times C$; F : 1.7, $P = 0.22$), intermediate ($T \times C$; F : 2.2, $P = 0.17$), and nonclassical monocytes ($T \times C$; F : 1.5, $P = 0.25$). In contrast to HWI_{pre} , the percentage of intermediate monocytes was not elevated following HWI_{post} (time; F : 3.4, $P = 0.06$; Table 3). There were no differences in the acute change between HWI_{pre} and HWI_{post} for total leukocyte ($T \times C$; F : 1.3, $P = 0.36$), monocyte ($T \times C$; F : 0.2, $P = 0.92$), lymphocyte ($T \times C$; F : 1.9, $P = 0.17$), and neutrophil ($T \times C$; F : 0.8, $P = 0.56$) numbers. Finally, the acute change in plasma nitrite concentration was similar between HWI_{pre} and HWI_{post} ($T \times C$; F : 1.3, $P = 0.30$; Fig. 3).

Chronic effects of the HWI intervention period. Table 4 shows the physiological responses during the HWI sessions of

the intervention period. Body mass did not change in INT following the intervention period (92.1 ± 9.2 to 92.3 ± 9.5 kg, F : 0.01, $P = 0.92$). Both systolic [time; F : 5.1, $P = 0.05$, ES: 0.60 (0.34–1.44)] and diastolic blood pressure [time; F : 14.3, $P = 0.003$, ES: 0.64 (0.32–1.47)] were lowered following the intervention period. Resting HR (time; F : 0.3, $P = 0.54$) and rectal temperature (time; F : 0.4, $P = 0.22$) were not affected by the intervention period (Table 2). Physical activity levels were not different from habitual physical activity (as reported at the start of the intervention period) during the intervention period (time; F : 0.2, $P = 0.64$).

The effect of the intervention period on resting IL-6, iHsp72, and eHsp72 levels is presented in Fig. 4. Resting levels of IL-6 and iHsp72 in total monocytes were not altered following the intervention period ($T \times G$; IL-6 F : 0.1, $P = 0.87$; iHsp72 F : 0.2, $P = 0.59$). The same was true for the expression of iHsp72 in the monocyte subsets ($T \times G$; classical monocytes F : 1.8, $P = 0.14$; intermediate monocytes F : 1.2, $P = 0.39$; nonclassical monocytes F : 0.3, $P = 0.78$). eHsp72 was lowered in INT compared with CON [difference in fold change between groups; F : 6.8; $P = 0.03$, ES: 1.00 (0.73–1.26)]. This resulted in a lower R in INT compared with CON [group; F : 6.0, $P = 0.04$, ES: 0.34 (0.21–0.51)]. The change in the distribution of

Table 3. Distribution of monocyte subsets in whole blood following hot water immersion and the control trial

	Classical Monocytes, %	Intermediate Monocytes, %	Nonclassical Monocytes, %
AMB			
Pre	94.6 $\pm$ 2.3	1.3 $\pm$ 0.4	2.6 $\pm$ 0.7
Post	94.0 $\pm$ 4.8	1.3 $\pm$ 0.3	2.4 $\pm$ 0.5
Post+2h	94.2 $\pm$ 1.9	1.4 $\pm$ 0.6	2.6 $\pm$ 1.1
HWI_{pre}			
Pre	94.50 $\pm$ 4.0	1.3 $\pm$ 0.6	2.2 $\pm$ 0.9
Post	91.2 $\pm$ 5.8	1.7 $\pm$ 0.7*	3.4 $\pm$ 1.9*
Post+2h	94.4 $\pm$ 2.7	1.3 $\pm$ 0.4	2.0 $\pm$ 0.4
HWI_{post}			
Pre	94.2 $\pm$ 1.2	1.5 $\pm$ 0.6	3.1 $\pm$ 1.0
Post	93.9 $\pm$ 1.7	1.4 $\pm$ 0.4†	3.5 $\pm$ 0.4*
Post+2h	94.7 $\pm$ 1.5	1.2 $\pm$ 0.5	2.7 $\pm$ 1.3

Values are means $\pm$ SD. AMB, control trial; HWI_{post} , hot water immersion (HWI) session following HWI intervention period; HWI_{pre} , HWI session before HWI intervention period; Post, after HWI or AMB; Post+2h, 2 h after HWI or AMB; Pre, before HWI or AMB. *Significantly different from AMB; †significant difference between HWI_{pre} and HWI_{post} .

Table 4. Physiological responses during the sessions of the 2-wk intervention period

Parameter	Sessions 1–5		Sessions 6–10	
	Pre	End	Pre	End
T_{tym} , °C	35.3 $\pm$ 0.4	37.5 $\pm$ 0.2*	35.1 $\pm$ 0.3	37.5 $\pm$ 0.3*
TS (1 to 9)	4.8 $\pm$ 0.5	6.6 $\pm$ 0.2*	4.9 $\pm$ 0.4	6.7 $\pm$ 0.2*
Basic affect (–5 to +5)	1.0 $\pm$ 1.0	0.0 $\pm$ 2.0	1.2 $\pm$ 1.6	–0.7 $\pm$ 1.8
HR, beats/min	67 $\pm$ 13	105 $\pm$ 2*	68 $\pm$ 14	105 $\pm$ 3*

Values are means $\pm$ SD. Data are means of 5 sessions during INT. Sessions 1–5 lasted 45 min, whereas sessions 6–10 lasted 60 min. End, measurement taken in the final 30 s of the session; HR, heart rate; Pre, before sessions; TS, thermal sensation; T_{tym} , tympanic temperature. *Significantly different from Pre.

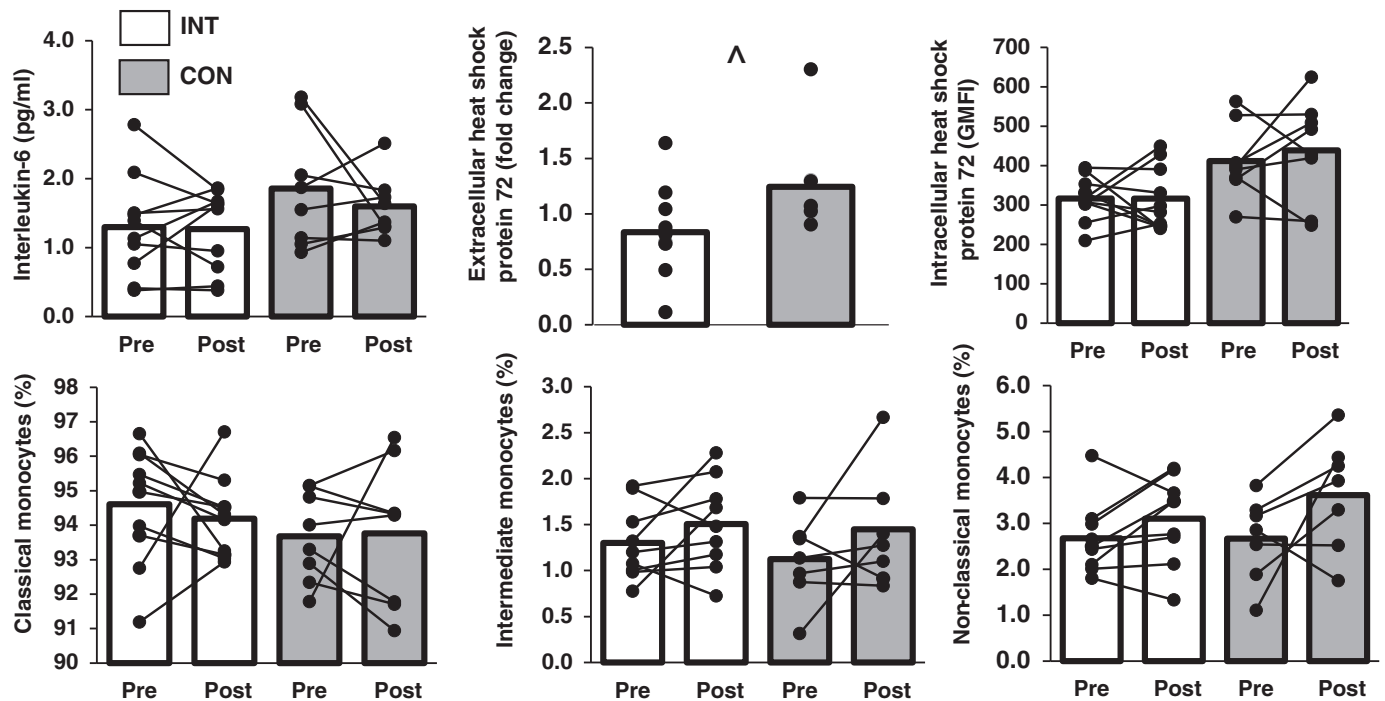


Fig. 4. Resting levels of the inflammatory outcome measures before and after the hot water immersion (HWI) intervention period. CON, control group ($n = 8$); GMFI, geometric mean fluorescence intensity; INT, intervention group ($n = 10$). Black lines represent individual data points, whereas bars represent the group mean. ^Significant difference between groups.

monocyte subsets in the circulation at rest was not different in INT compared with CON ($T \times G$; classical monocytes $F: 0.8$, $P = 0.52$; intermediate monocytes $F: 1.1$, $P = 0.23$; nonclassical monocytes $F: 1.8$, $P = 0.14$; Fig. 4).

Fasting blood glucose concentrations were lower in INT compared with CON following the intervention period [$T \times G$; $F: 5.0$, $P = 0.04$, ES: 0.68 (0.42–0.97); Fig. 5]. Fasting insulin concentrations did not change in INT compared with CON [$T \times G$; $F: 1.3$, $P = 0.30$, ES: 0.50 (–0.46 to 1.42)]. However, following inspection of the individual data, an outlier was detected (Fig. 5, right, gray line), which was confirmed using the methods for outlier detection postulated by Leys et al. (46). After removing the insulin data of this participant, there was a larger decrease in fasting insulin in INT compared with CON [$T \times G$; $F: 4.8$, $P = 0.04$, ES: 1.06 (0.02–2.00)]. HOMA-IR was also reduced to a larger extent in INT compared with CON [$T \times G$; $F: 5.5$, $P = 0.03$, ES: 1.07 (0.08–2.06)]. Finally, there was no difference in the change of resting plasma nitrite concentrations between INT and CON (INT 321 ± 69 to

234 ± 64 nM; CON 230 ± 57 to 262 ± 77 nM; $T \times G$; $F: 1.7$, $P = 0.17$).

One week following the Post blood sample, resting iHsp72 (Pre: 307 ± 53 GMFI, Post: 309 ± 69 GMFI, Post+1wk: 358 ± 116 GMFI; time; $F: 1.8$, $P = 0.22$), IL-6 (Pre: 1.22 ± 0.52 pg/ml, Post: 1.31 ± 0.53 pg/ml, Post+1wk: 1.12 ± 0.65 pg/ml; time; $F: 0.2$, $P = 0.67$), the percentage of classical monocytes (Pre: $94.4 \pm 1.8\%$, Post: $91.9 \pm 4.5\%$, Post+1wk: $94.1 \pm 1.3\%$; time; $F: 1.7$, $P = 0.18$), intermediate monocytes (Pre: $1.25 \pm 0.38\%$, Post: $1.69 \pm 0.73\%$, Post+1wk: $1.47 \pm 0.51\%$; time; $F: 1.0$, $P = 0.27$), and nonclassical monocytes (Pre: $2.70 \pm 0.92\%$, Post: $3.10 \pm 1.09\%$, Post+1wk: $3.39 \pm 1.35\%$; time; $F: 1.0$, $P = 0.28$) were not changed compared with either Pre or Post. Resting concentrations of eHsp72 were elevated compared with Post [fold change Pre-Post: 0.83 ± 0.41 , fold change Pre-Post+1wk: 1.28 ± 0.34 , time; $F: 5.8$, $P = 0.03$, ES: 0.83 (0.20–1.84)]. The lowering of fasting blood glucose following the intervention period was still present at Post+1wk [Pre: 4.44 ± 0.93 mmol/l, Post:

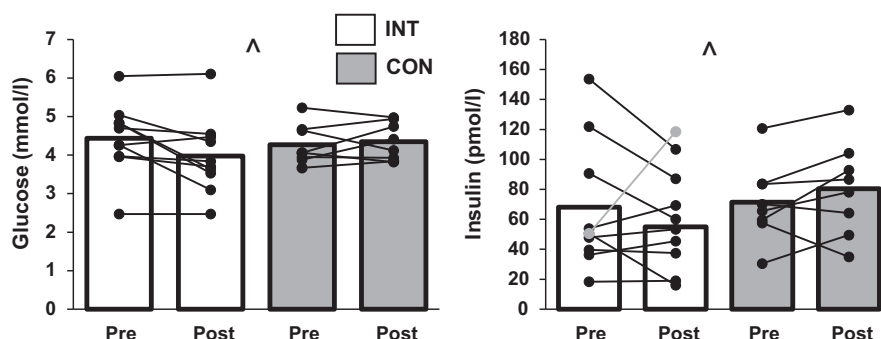


Fig. 5. Fasting blood glucose and plasma insulin concentrations for the intervention and control groups. CON, control group ($n = 8$); INT, intervention group ($n = 10$). Black lines represent individual data points, whereas bars represent the group mean. Participant with gray line does not contribute to the bar representing the group mean. ^Significant time $\times$ group interaction.

3.98 ± 0.98 mmol/l, Post+1wk: 3.89 ± 0.77 mmol/l, time; F : 25.1, $P = 0.001$, ES: 0.61 (0.08–1.32)]. However, fasting insulin was elevated at Post+1wk compared with Post [Pre: 68.10 ± 44.65 pmol/l, Post: 51.7 ± 27.3 pmol/l, Post+1wk: 72.6 ± 56.3 pmol/l, time; F : 4.5, $P = 0.05$, ES: 0.53 (0.05–1.08)], returning to the insulin concentrations found before the intervention (Pre-Post+1wk, time; F : 1.1, $P = 0.21$). There was no difference in HOMA-IR at Post+1wk compared with Post (Pre: 13.91 ± 11.09 , Post: 8.99 ± 7.89 , Post+1wk: 12.40 ± 10.01 , time; F : 4.1, $P = 0.06$) or Pre (time; F : 0.8, $P = 0.47$). Plasma nitrite concentrations were not changed at Post+1wk compared with Pre or Post (Pre: 314 ± 61 nM, Post: 247 ± 66 nM, Post+1wk: 304 ± 91 nM; time; F : 3.9, $P = 0.09$).

Correlations. During HWI_{pre} there was no correlation between the peak core temperature attained and the acute change in iHsp72 expression ($r = -0.11$, $P = 0.77$), plasma IL-6 concentration ($r = 0.23$, $P = 0.55$), or nitrite concentration ($r = 0.04$, $P = 0.91$). Following the chronic intervention, there was a negative correlation between plasma insulin concentration at baseline and its change following the intervention ($r = -0.45$, $P = 0.01$). There was no relationship with insulin at baseline and the change in blood glucose concentrations ($r = 0.23$, $P = 0.33$). No correlation was observed between baseline blood glucose concentration and the chronic change in insulin ($r = -0.28$, $P = 0.27$) or glucose concentrations ($r = 0.29$, $P = 0.25$). In addition, there was no correlation between the fold change in eHsp72 following the intervention and the change in insulin ($r = 0.61$, $P = 0.06$) or glucose concentrations ($r = 0.03$, $P = 0.94$). Finally, there was no correlation between the chronic change in iHsp72 expression and the chronic change in insulin ($r = -0.16$, $P = 0.66$) or glucose concentrations ($r = 0.21$, $P = 0.56$).

DISCUSSION

This study investigated the acute inflammatory response to HWI as well as the potential of chronic HWI to improve inflammatory and metabolic profiles at rest. Acute HWI evoked elevated plasma IL-6 and nitrite concentrations and an increase in the percentage of intermediate and nonclassical monocytes. This was, however, not accompanied by an increase in iHsp72 expression. Two weeks of chronic HWI reduced fasting glucose, insulin, and eHsp72 concentrations. Together, this indicates that HWI may be a useful strategy to improve aspects of the inflammatory profile and glucose metabolism in individuals without the physical capacity to do so using exercise training.

Acute responses to HWI. Our observation that 1 h of HWI in water set at 39°C induced a significant increase in plasma IL-6 concentrations corroborates the notion that increases in body temperature can serve as an independent stressor to induce an acute inflammatory response. Previous studies employing 1 h of HWI have shown increases in plasma IL-6 concentrations comparable to those of the present study (16, 45), whereas 2 h of HWI results in a more marked IL-6 response (43). Consistent with exercise studies (17), this suggests that the IL-6 response to HWI is dose dependent. In line with this, a more intense HWI protocol than used in the present study (i.e., longer duration or warmer water) may be required to induce changes in iHsp72 or eHsp72. Oehler et al. (54) reported an

acute increase in iHsp72 following HWI of 2 h in water set at 39.5°C, whereas a session of 1 h did not result in elevated iHsp72 expression (50). On the other hand, Faulkner et al. (16) reported acute increases in eHsp72 following immersion up to the waistline for 1 h in water set at 40°C, resulting in a ~1°C increase in core temperature. As the acute inflammatory response to HWI seems dose dependent, it is conceivable that there may exist a threshold in core or muscle temperature or time accrued above this threshold that needs to be reached to induce an iHsp72 response. Using exercise as a stressor, Gibson et al. (24) have suggested that at least ~27 min above a core temperature of 38.5°C are needed to induce the upregulation of Hsp72 mRNA. In the present study, participants' rectal temperature exceeded 38.5°C for ~15 min only. This may also explain why an acute increase in iHsp72 following passive heating is a consistent finding in animal studies (28, 64), but not in human studies (50), as the endogenous heat stress imposed in the former is much higher compared with the present and other studies in humans. Of note, the required heat stress might need to be even higher to induce acute increases in circulating eHsp72 concentrations (23).

Although the HWI protocol used in this study did not elevate iHsp72 expression, the acute increase in IL-6 concentrations indicates that in analogy to exercise, passive heating can also induce an acute inflammatory response, possibly leading to the circulating anti-inflammatory milieu postulated by Petersen and Pedersen as one of the benefits of exercise (57). Although it is now widely acknowledged that contracting skeletal muscle is the main source of IL-6 during acute exercise (17), it is not clear whether this is also the case for HWI. However, skeletal muscle is suggested to secrete IL-6 in response to increases in local temperature (66). HWI for 1 h in water set at 40°C leads indeed to a muscle temperature increase of ~2.5°C (16). Suggested mechanisms for the acute inflammatory response following passive heating are the influx of calcium via the opening of the thermosensitive transient receptor potential 1 (53) and the activation of heat shock factor 1, which can both result in the production of IL-6 and Hsp72 (66). In addition, circulating monocytes are potent producers of cytokines and might be a source of IL-6 found in the circulation following HWI (1). The acute recruitment of intermediate and nonclassical monocytes seen following HWI in this study could indeed have led to increased IL-6 secretion into the circulation as these subsets are known to release more IL-6 in response to an *in vitro* stimulant such as lipopolysaccharide (30). However, since monocytes only represent a small percentage of leukocytes, it is not known what the impact of acute changes in circulating monocyte subsets on circulating cytokines is (65). Nevertheless, since the proportion of relatively inflammatory monocytes (i.e., intermediate and nonclassical monocytes) at rest is positively associated with the risk for a range of chronic diseases (69), the acute shift following HWI found in this study provides rationale for further research into the potential of HWI interventions to chronically alter the distribution of monocyte subsets in the circulation.

Although the interest in HWI to reduce chronic low-grade inflammation is a relatively recent phenomenon, its potential to increase blood flow and enhance vascular function is more established (13). Nevertheless, we show for the first time an acute increase in the bioavailability of the vasodilator NO in response to HWI in humans, possibly mediated by the en-

hanced activation of endothelial NOS in response to the increase in shear stress and/or local temperature (19). Additionally, as Hsp90 acts as an agonist for NO production by endothelial NOS, the acute increase in NO bioavailability may have been mediated by an increased expression of Hsp90 (22). Future studies are therefore needed to identify the potential of HWI to increase Hsp90 expression. Since the acute increase in NO following HWI has the potential to aid tissue blood flow and is implicated in the translocation of glucose transporter isoform 4 (GLUT4) to the plasma membrane of skeletal muscle cells during exercise (59), HWI has the potential to facilitate glucose disposal in skeletal muscle and other tissues (2, 20). In support, animal studies suggest GLUT4 translocation (25) and enhanced insulin sensitivity in skeletal muscle (27) following an acute HWI session. Of note, in the present study the acute effects of HWI on glucose disposal were not assessed, and the implications of an acute increase in NO bioavailability on glucose disposal are therefore only speculative. Indeed, the chronic reduction in fasting glucose and insulin found in the present study occurred independently of changes in resting plasma nitrite concentrations.

If passive heating is to be successfully introduced as a health-promoting intervention in practice, it is important to assess perceptual responses to provide insight into its potential to influence adherence rates to the intervention (68). In the present study, the perceptual responses during 1 h of HWI indicated profound feelings of discomfort similar to those reported during high-intensity interval training (32, 38). This implies that further increases in water temperature or session duration would result in an activity that is difficult to adhere to (15). Therefore, although more intense HWI sessions than the one used in the present study seem to be needed to induce an acute Hsp72 response, the practical application of HWI sessions such as the one applied in the study of Oehler et al. (54; 2 h at 39.5°C) in the general population is questionable. Moreover, the absence of more positive affective responses during HWI_{post} compared with HWI_{pre} suggests that no short-term improvements in the perceptual responses can be expected as a result of regular engagement in HWI. Therefore, future studies could test different HWI protocols in an attempt to optimize the balance between delivering a HWI stimulus that evokes the necessary inflammatory and metabolic benefits without eliciting negative affective responses that have the potential to limit adherence to the intervention. Finally, although HWI did not induce acute changes in Hsp72, we did observe acute elevations of nitrite and IL-6 in addition to chronic improvements in fasting glucose, insulin, and eHsp72. This suggests that there may be no need to further increase the thermal load of the HWI sessions to improve metabolic health and that the focus could be directed toward the improvement of the perceptual responses during HWI. A titration study in which the thermal load is gradually reduced may be useful to gain insight into the minimal passive heat stress needed to induce acute changes in factors such as plasma IL-6 concentrations and NO bioavailability and its impact on the perceptual responses during HWI.

Chronic effects of the HWI intervention period. As suggested by several authors (34, 42, 48), HWI interventions could serve as a strategy to improve insulin sensitivity, possibly via the elevation of iHsp72 expression and/or reduced chronic low-grade inflammation. In line with this suggestion, which

emanated from the pilot study by Hooper (33), fasting glucose and insulin concentrations were reduced following the 2-wk HWI intervention period applied in the present study. This was accompanied by a reduction in eHsp72 concentrations. However, no changes in resting iHsp72 expression or plasma IL-6 and nitrite concentrations were found.

Animal studies suggest increased basal iHsp72 levels as a mechanism behind the beneficial changes in insulin sensitivity reported following HWI (6, 26). Moreover, Hsp72 knockout mice are highly insulin resistant and do not experience similar benefits from passive heating strategies compared with mice expressing Hsp72 (12). However, in the present study, reductions in fasting glucose and insulin were found in the absence of changes in iHsp72. The reason for this discrepancy might lie in the tissue in which iHsp72 was assessed. Whereas most animal studies have investigated iHsp72 in skeletal muscle, in the present study, iHsp72 expression was assessed in monocytes. Although the acute iHsp72 responses in leukocytes follow the same pattern as those found in skeletal muscle (64) and *in vitro* heat shock upregulates iHsp72 expression in monocytes (62), the chronic adaptations to heat therapy and health interventions in monocytes are less clear. Whereas heat acclimation using exercise can induce increases in monocyte iHsp72 expression (44), trained runners actually express lower levels of iHsp72 in leukocytes compared with their sedentary counterparts (18). More studies that simultaneously measure iHsp72 expression in both tissues following health interventions are therefore needed to resolve the mechanisms for enhanced glucose metabolism after HWI. It should be acknowledged that the chronic intervention may have impacted on other factors implicated in glucose metabolism, as, for instance, passive lower limb heating can chronically elevate peroxisome proliferator-activated receptor- γ coactivator-1 α (PGC-1 α) expression (28).

Despite no changes in resting iHsp72, eHsp72 concentrations were significantly lowered following the intervention period. When present in the circulation, eHsp72 can activate monocytes via the Toll-like receptor 4/CD14 complex, resulting in the secretion of proinflammatory cytokines such as IL-6, tumor necrosis factor- α (TNF- α), and IL- β (1). As the latter cytokines can directly interfere with insulin sensitivity (35), it is suggested that the deleterious effects of eHsp72 on health are exhibited via this mechanism (37). Additionally, the positive change in *R* in INT might be indicative of an improved inflammatory profile following the intervention period, as suggested by Krause et al. (41). However, the influence of eHsp72 and changes in *R* on glucose metabolism needs to be studied in more detail.

Whereas previous studies have found changes in the inflammatory profile following short-term health interventions, the relatively short duration of the HWI intervention period might have been the reason for the absence of changes in resting levels of iHsp72, IL-6, monocyte subset distribution, and NO bioavailability. On the other hand, it is striking that only 10 HWI sessions resulted in reductions in fasting glucose, insulin, and blood pressure in men that were sedentary and overweight but did not show signs of prediabetes or strongly elevated inflammatory markers at baseline. The positive correlation between baseline fasting insulin concentrations and the reduction in fasting insulin following the intervention suggests that those with more impaired metabolic health might benefit most

from HWI. The lowered blood pressure following the intervention period supports recent findings by Brunt et al. (4) suggesting that HWI may also be a potent strategy to improve vascular health. Although iHsp72- and NO-mediated mechanisms are suggested to play a role in this effect (5), the improvements in blood pressure in the present study were independent of changes in resting levels of both measures.

Altogether, the present study provides a strong rationale to pursue further research on the potential of passive heating strategies to enhance (cardio)metabolic health. For instance, future studies should consider using more robust measures of insulin sensitivity (e.g., oral glucose tolerance testing), implementing longer-term interventions, and exploring its effectiveness and feasibility in populations that could benefit most from this alternative health intervention (e.g., individuals with a spinal cord injury, frail elderly, or those with other conditions that interfere with exercise participation). Additionally, future studies in humans are needed to clarify the role of inflammatory markers in glucose metabolism. In this regard, the relatively modest heat stress imposed in the present study may be considered a limitation. Although here an applicable model of passive heating is presented, future mechanistic studies may consider increasing body temperature to a larger extent and for longer durations. For instance, a passive heating model that is more likely to elevate iHsp72 expression may aid our understanding of the importance of this marker for glucose metabolism in humans. Finally, although there was no acute iHsp72 response following HWI and resting iHsp72 expression in monocytes was not changed following the intervention, an elevated iHsp72 expression in skeletal muscle for up to 7 days has been reported following exercise (51). Therefore, the resting and postimmersion inflammatory and metabolic markers may have been influenced by the potentially elevated iHsp72 expression in skeletal muscle.

In summary, a single HWI session induces an acute inflammatory response, indicated by acute elevations in IL-6, changes in the monocyte subset distribution, and increase in NO synthesis, indicated by increased plasma nitrite concentrations. However, these responses were not accompanied by acute increases in iHsp72 or eHsp72. The 2-wk HWI intervention period reduced fasting glucose and insulin, concomitant with lower resting eHsp72 concentrations but independent of iHsp72 expression and plasma IL-6 and nitrite concentrations at rest, as the latter markers did not change following the chronic intervention. Therefore, this study provides support for the use of HWI to improve aspects of the inflammatory profile and enhance glucose metabolism in sedentary, overweight men and might have implications for improving metabolic health in populations unable to meet the present physical activity recommendations.

ACKNOWLEDGMENTS

We thank Bryany Cornish, Ben Lowry, and Tim Lam for their excellent assistance in the process of data collection. Most of all, we are grateful for the effort and time the participants have dedicated to this research project.

GRANTS

We acknowledge the support of the National Institute for Health Research Leicester Biomedical Research Centre.

DISCLAIMERS

The views expressed are those of the authors and not necessarily those of the National Health Service, the National Institute for Health Research or the Department of Health.

DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

S.P.H., N.C.B., S.H.F., and C.A.L. conceived and designed research; S.P.H. performed experiments; S.P.H. and S.J.B. analyzed data; S.P.H., N.C.B., S.H.F., S.J.B., and C.A.L. interpreted results of experiments; S.P.H. prepared figures; S.P.H. drafted manuscript; S.P.H., N.C.B., S.H.F., S.J.B., and C.A.L. edited and revised manuscript; S.P.H., N.C.B., S.H.F., S.J.B., and C.A.L. approved final version of manuscript.

REFERENCES

1. Asea A, Rehli M, Kabingu E, Boch JA, Baré O, Auron PE, Stevenson MA, Calderwood SK. Novel signal transduction pathway utilized by extracellular HSP70: role of Toll-like receptor (TLR) 2 and TLR4. *J Biol Chem* 277: 15028–15034, 2002. doi:10.1074/jbc.M200497200.
2. Baron AD, Steinberg H, Brechtel G, Johnson A. Skeletal muscle blood flow independently modulates insulin-mediated glucose uptake. *Am J Physiol Endocrinol Metab* 266: E248–E253, 1994. doi:10.1152/ajpendo.1994.266.2.E248.
3. Brunt VE, Howard MJ, Francisco MA, Ely BR, Minson CT. Passive heat therapy improves endothelial function, arterial stiffness and blood pressure in sedentary humans. *J Physiol* 594: 5329–5342, 2016. doi:10.1113/JP272453.
4. Brunt VE, Howard MJ, Francisco MA, Ely BR, Minson CT. Reply from Vienna E. Brunt, Matthew J. Howard, Michael A. Francisco, Brett R. Ely and Christopher T. Minson. *J Physiol* 595: 3669–3670, 2017. doi:10.1113/JP274129.
5. Chung J, Nguyen A-K, Henstridge DC, Holmes AG, Chan MHS, Mesa JL, Lancaster GI, Southgate RJ, Bruce CR, Duffy SJ, Horvath I, Mestrlil R, Watt MJ, Hooper PL, Kingwell BA, Vigh L, Hevener A, Febbraio MA. HSP72 protects against obesity-induced insulin resistance. *Proc Natl Acad Sci USA* 105: 1739–1744, 2008. doi:10.1073/pnas.0705799105.
6. Cohen J. A power primer. *Psychol Bull* 112: 155–159, 1992. doi:10.1037/0033-2909.112.1.155.
7. Craig CL, Marshall AL, Sjöström M, Bauman AE, Booth ML, Ainsworth BE, Pratt M, Ekelund U, Yngve A, Sallis JF, Oja P. International Physical Activity Questionnaire: 12-country reliability and validity. *Med Sci Sports Exerc* 35: 1381–1395, 2003. doi:10.1249/01.MSS.0000078924.61453.FB.
8. Dandona P, Ajlada A, Bandyopadhyay A. Inflammation: the link between insulin resistance, obesity and diabetes. *Trends Immunol* 25: 4–7, 2004. doi:10.1016/j.it.2003.10.013.
9. Dill DB, Costill DL. Calculation of percentage changes in volumes of blood, plasma, and red cells in dehydration. *J Appl Physiol* 37: 247–248, 1974. doi:10.1152/jappl.1974.37.2.247.
10. Dorsey CM, Lukas SE, Teicher MH, Harper D, Winkelman JW, Cunningham SL, Satlin A. Effects of passive body heating on the sleep of older female insomniacs. *J Geriatr Psychiatry Neurol* 9: 83–90, 1996. doi:10.1177/089198879600900203.
11. Drew BG, Ribas V, Le JA, Henstridge DC, Phun J, Zhou Z, Soleymani T, Daraei P, Sitz D, Vergnes L, Wanagat J, Reue K, Febbraio MA, Hevener AL. HSP72 is a mitochondrial stress sensor critical for Parkin action, oxidative metabolism, and insulin sensitivity in skeletal muscle. *Diabetes* 63: 1488–1505, 2014. doi:10.2337/db13-0665.
12. Drummond PD. Thermoregulatory response to passive body heating in borderline hypertension. *Clin Auton Res* 3: 233–238, 1993. doi:10.1007/BF01829011.
13. Durnin JV, Womersley J. Body fat assessed from total body density and its estimation from skinfold thickness: measurements on 481 men and women aged from 16 to 72 years. *Br J Nutr* 32: 77–97, 1974. doi:10.1079/BJN19740060.
14. Ekkekakis P, Parfitt G, Petruzzello SJ. The pleasure and displeasure people feel when they exercise at different intensities: decennial update and progress towards a tripartite rationale for exercise intensity prescription.

- tion. *Sports Med* 41: 641–671, 2011. doi:10.2165/11590680-000000000-00000.
16. Faulkner SH, Jackson S, Fatania G, Leicht CA. The effect of passive heating on heat shock protein 70 and interleukin-6: a possible treatment tool for metabolic diseases? *Temperature (Austin)* 4: 292–304, 2017. doi:10.1080/23328940.2017.1288688.
 17. Febbraio MA, Pedersen BK. Contraction-induced myokine production and release: is skeletal muscle an endocrine organ? *Exerc Sport Sci Rev* 33: 114–119, 2005. doi:10.1097/00003677-200507000-00003.
 18. Fehrenbach E, Passek F, Niess AM, Pohla H, Weinstock C, Dickhuth HH, Northoff H. HSP expression in human leukocytes is modulated by endurance exercise. *Med Sci Sports Exerc* 32: 592–600, 2000. doi:10.1097/00005768-200003000-00007.
 19. Förstermann U, Sessa WC. Nitric oxide synthases: regulation and function. *Eur Heart J* 33: 829–837, 2012. doi:10.1093/eurheartj/ehs304.
 20. Franks PW, Luan J, Barroso I, Brage S, Gonzalez Sanchez JL, Ekelund U, Ríos MS, Schafer AJ, O'Rahilly S, Wareham NJ. Variation in the eNOS gene modifies the association between total energy expenditure and glucose intolerance. *Diabetes* 54: 2795–2801, 2005. doi:10.2337/diabetes.54.9.2795.
 21. Gagge AP, Stolwijk JA, Hardy JD. Comfort and thermal sensations and associated physiological responses at various ambient temperatures. *Environ Res* 1: 1–20, 1967. doi:10.1016/0013-9351(67)90002-3.
 22. García-Cardeña G, Fan R, Shah V, Sorrentino R, Cirino G, Papadopoulos A, Sessa WC. Dynamic activation of endothelial nitric oxide synthase by Hsp90. *Nature* 392: 821–824, 1998. doi:10.1038/33934.
 23. Gibson OR, Dennis A, Parfitt T, Taylor L, Watt PW, Maxwell NS. Extracellular Hsp72 concentration relates to a minimum endogenous criteria during acute exercise-heat exposure. *Cell Stress Chaperones* 19: 389–400, 2014. doi:10.1007/s12192-013-0468-1.
 24. Gibson OR, Tuttle JA, Watt PW, Maxwell NS, Taylor L. Hsp72 and Hsp90α mRNA transcription is characterised by large, sustained changes in core temperature during heat acclimation. *Cell Stress Chaperones* 21: 1021–1035, 2016. doi:10.1007/s12192-016-0726-0.
 25. Goto A, Egawa T, Sakon I, Oshima R, Ito K, Serizawa Y, Sekine K, Tsuda S, Goto K, Hayashi T. Heat stress acutely activates insulin-independent glucose transport and 5'-AMP-activated protein kinase prior to an increase in HSP72 protein in rat skeletal muscle. *Physiol Rep* 3: e12601, 2015. doi:10.14814/phy2.12601.
 26. Gupte AA, Bomhoff GL, Swerdlow RH, Geiger PC. Heat treatment improves glucose tolerance and prevents skeletal muscle insulin resistance in rats fed a high-fat diet. *Diabetes* 58: 567–578, 2009. doi:10.2337/db08-1070.
 27. Gupte AA, Bomhoff GL, Touchberry CD, Geiger PC. Acute heat treatment improves insulin-stimulated glucose uptake in aged skeletal muscle. *J Appl Physiol* (1985) 110: 451–457, 2011. doi:10.1152/jappphysiol.00849.2010.
 28. Hafen PS, Preece CN, Sorensen JR, Hancock CR, Hyldahl RD. Repeated exposure to heat stress induces mitochondrial adaptation in human skeletal muscle. *J Appl Physiol* (1985) 125: 1447–1455, 2018. doi:10.1152/jappphysiol.00383.2018.
 29. Hawley JA, Lessard SJ. Exercise training-induced improvements in insulin action. *Acta Physiol (Oxf)* 192: 127–135, 2008. doi:10.1111/j.1748-1716.2007.01783.x.
 30. Heine GH, Ortiz A, Massy ZA, Lindholm B, Wiecek A, Martínez-Castelao A, Covic A, Goldsmith D, Süleymanlar G, London GM, Parati G, Sicari R, Zoccali C, Fliser D; European Renal and Cardiovascular Medicine (EURECA-m) Working Group of the European Renal Association-European Dialysis and Transplant Association (ERA-EDTA). Monocyte subpopulations and cardiovascular risk in chronic kidney disease. *Nat Rev Nephrol* 8: 362–369, 2012. doi:10.1038/nrneph.2012.41.
 31. Henstridge DC, Whitham M, Febbraio MA. Chaperoning to the metabolic party: the emerging therapeutic role of heat-shock proteins in obesity and type 2 diabetes. *Mol Metab* 3: 781–793, 2014. doi:10.1016/j.molmet.2014.08.003.
 32. Hoekstra SP, Bishop NC, Leicht CA. Can intervals enhance the inflammatory response and enjoyment in upper-body exercise? *Eur J Appl Physiol* 117: 1155–1163, 2017. doi:10.1007/s00421-017-3602-4.
 33. Hooper PL. Hot-tub therapy for type 2 diabetes mellitus. *N Engl J Med* 341: 924–925, 1999. doi:10.1056/NEJM199909163411216.
 34. Hooper PL, Balogh G, Rivas E, Kavanagh K, Vigh L. The importance of the cellular stress response in the pathogenesis and treatment of type 2 diabetes. *Cell Stress Chaperones* 19: 447–464, 2014. doi:10.1007/s12192-014-0493-8.
 35. Hotamisligil GS. Inflammatory pathways and insulin action. *Int J Obes Relat Metab Disord* 27, Suppl 3: S53–S55, 2003. doi:10.1038/sj.ijo.0802502.
 36. Ives SJ, Andthacka RH, Kwon SH, Shiu YT, Ruan T, Noyes RD, Zhang QJ, Symons JD, Richardson RS. Heat and α₁-adrenergic responsiveness in human skeletal muscle feed arteries: the role of nitric oxide. *J Appl Physiol* (1985) 113: 1690–1698, 2012. doi:10.1152/jappphysiol.00955.2012.
 37. Johnson JD, Fleshner M. Releasing signals, secretory pathways, and immune function of endogenous extracellular heat shock protein 72. *J Leukoc Biol* 79: 425–434, 2006. doi:10.1189/jlb.0905523.
 38. Jung ME, Bourne JE, Little JP. Where does HIT fit? An examination of the affective response to high-intensity intervals in comparison to continuous moderate- and continuous vigorous-intensity exercise in the exercise intensity-affect continuum. *PLoS One* 9: e114541, 2014. doi:10.1371/journal.pone.0114541.
 39. Karakas M, Koenig W. CRP in cardiovascular disease. *Herz* 34: 607–613, 2009. doi:10.1007/s00059-009-3305-7.
 40. Krause M, Heck TG, Bittencourt A, Scamazzon SP, Newsholme P, Curi R, Homem de Bittencourt PI Jr. The chaperone balance hypothesis: the importance of the extracellular to intracellular HSP70 ratio to inflammation-driven type 2 diabetes, the effect of exercise, and the implications for clinical management. *Mediators Inflamm* 2015: 249205, 2015. doi:10.1155/2015/249205.
 41. Krause M, Ludwig MS, Heck TG, Takahashi HK. Heat shock proteins and heat therapy for type 2 diabetes: pros and cons. *Curr Opin Clin Nutr Metab Care* 18: 374–380, 2015. doi:10.1097/MCO.0000000000000183.
 42. Laing SJ, Jackson AR, Walters R, Lloyd-Jones E, Whitham M, Maassen N, Walsh NP. Human blood neutrophil responses to prolonged exercise with and without a thermal clamp. *J Appl Physiol* (1985) 104: 20–26, 2008. doi:10.1152/jappphysiol.00792.2007.
 43. Lee BJ, Miller A, James RS, Thake CD. Cross acclimation between heat and hypoxia: heat acclimation improves cellular tolerance and exercise performance in acute normobaric hypoxia. *Front Physiol* 7: 78, 2016. doi:10.3389/fphys.2016.00078.
 44. Leicht CA, Kouda K, Umamoto Y, Banno M, Kinoshita T, Moriki T, Nakamura T, Bishop NC, Goosey-Tolfrey VL, Tajima F. Hot water immersion induces an acute cytokine response in cervical spinal cord injury. *Eur J Appl Physiol* 115: 2243–2252, 2015. doi:10.1007/s00421-015-3206-9.
 45. Leys C, Ley C, Klein O, Bernard P, Licata L. Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median. *J Exp Soc Psychol* 49: 764–766, 2013. doi:10.1016/j.jesp.2013.03.013.
 46. Matthews DR, Hosker JP, Rudenski AS, Naylor BA, Treacher DF, Turner RC. Homeostasis model assessment: insulin resistance and β-cell function from fasting plasma glucose and insulin concentrations in man. *Diabetologia* 28: 412–419, 1985. doi:10.1007/BF00280883.
 47. McCarty MF, Barroso-Aranda J, Contreras F. Regular thermal therapy may promote insulin sensitivity while boosting expression of endothelial nitric oxide synthase: effects comparable to those of exercise training. *Med Hypotheses* 73: 103–105, 2009. doi:10.1016/j.mehy.2008.12.020.
 48. Morton JP, MacLaren DP, Cable NT, Campbell IT, Evans L, Bongers T, Griffiths RD, Kayani AC, McArdle A, Drust B. Elevated core and muscle temperature to levels comparable to exercise do not increase heat shock protein content of skeletal muscle of physically active men. *Acta Physiol (Oxf)* 190: 319–327, 2007. doi:10.1111/j.1748-1716.2007.01711.x.
 49. Morton JP, MacLaren DP, Cable NT, Bongers T, Griffiths RD, Campbell IT, Evans L, Kayani A, McArdle A, Drust B. Time course and differential responses of the major heat shock protein families in human skeletal muscle following acute nondamaging treadmill exercise. *J Appl Physiol* (1985) 101: 176–182, 2006. doi:10.1152/jappphysiol.00046.2006.
 50. Nowotny B, Nowotny PJ, Strassburger K, Roden M. Precision and accuracy of blood glucose measurements using three different instruments. *Diabet Med* 29: 260–265, 2012. doi:10.1111/j.1464-5491.2011.03406.x.
 51. Obi S, Nakajima T, Hasegawa T, Kikuchi H, Oguri G, Takahashi M, Nakamura F, Yamasoba T, Sakuma M, Toyoda S, Tei C, Inoue T. Heat induces interleukin-6 in skeletal muscle cells via TRPV1/PKC/CREB pathways. *J Appl Physiol* (1985) 122: 683–694, 2017. doi:10.1152/jappphysiol.00139.2016.

54. Oehler R, Pusch E, Zellner M, Dungal P, Hergovics N, Homoncik M, Eliassen MM, Brabec M, Roth E. Cell type-specific variations in the induction of hsp70 in human leukocytes by feverlike whole body hyperthermia. *Cell Stress Chaperones* 6: 306–315, 2001. doi:[10.1379/1466-1268\(2001\)006<0306:CTSVIT>2.0.CO;2](https://doi.org/10.1379/1466-1268(2001)006<0306:CTSVIT>2.0.CO;2).
55. Oyama J, Kudo Y, Maeda T, Node K, Makino N. Hyperthermia by bathing in a hot spring improves cardiovascular functions and reduces the production of inflammatory cytokines in patients with chronic heart failure. *Heart Vessels* 28: 173–178, 2013. doi:[10.1007/s00380-011-0220-7](https://doi.org/10.1007/s00380-011-0220-7).
56. Perneger TV. What's wrong with Bonferroni adjustments. *BMJ* 316: 1236–1238, 1998. doi:[10.1136/bmj.316.7139.1236](https://doi.org/10.1136/bmj.316.7139.1236).
57. Petersen AM, Pedersen BK. The anti-inflammatory effect of exercise. *J Appl Physiol* (1985) 98: 1154–1162, 2005. doi:[10.1152/jappphysiol.00164.2004](https://doi.org/10.1152/jappphysiol.00164.2004).
58. Raubenheimer K, Hickey D, Leveritt M, Fassett R, Ortiz de Zavallos Munoz J, Allen JD, Briskey D, Parker TJ, Kerr G, Peake JM, Pecheniuk NM, Neubauer O. Acute effects of nitrate-rich beetroot juice on blood pressure, hemostasis and vascular inflammation markers in healthy older adults: a randomized, placebo-controlled crossover study. *Nutrients* 9: 1270–1289, 2017. doi:[10.3390/nu9111270](https://doi.org/10.3390/nu9111270).
59. Roberts CK, Barnard RJ, Scheck SH, Balon TW. Exercise-stimulated glucose transport in skeletal muscle is nitric oxide dependent. *Am J Physiol Endocrinol Metab* 273: E220–E225, 1997. doi:[10.1152/ajpendo.1997.273.1.E220](https://doi.org/10.1152/ajpendo.1997.273.1.E220).
60. Sansbury BE, Hill BG. Regulation of obesity and insulin resistance by nitric oxide. *Free Radic Biol Med* 73: 383–399, 2014. doi:[10.1016/j.freeradbiomed.2014.05.016](https://doi.org/10.1016/j.freeradbiomed.2014.05.016).
61. Silverstein MG, Ordanes D, Wylie AT, Files DC, Milligan C, Presley TD, Kavanagh K. Inducing muscle heat shock protein 70 improves insulin sensitivity and muscular performance in aged mice. *J Gerontol A Biol Sci Med Sci* 70: 800–808, 2015. doi:[10.1093/gerona/glu119](https://doi.org/10.1093/gerona/glu119).
62. Simar D, Jacques A, Caillaud C. Heat shock proteins induction reduces stress kinases activation, potentially improving insulin signalling in monocytes from obese subjects. *Cell Stress Chaperones* 17: 615–621, 2012. doi:[10.1007/s12192-012-0336-4](https://doi.org/10.1007/s12192-012-0336-4).
63. Strokov IA, Manukhina EB, Bakhtina LY, Malyshev IY, Zoloev GK, Kazikhanova SI, Ametov AS. The function of endogenous protective systems in patients with insulin-dependent diabetes mellitus and polyneuropathy: effect of antioxidant therapy. *Bull Exp Biol Med* 130: 986–990, 2000. doi:[10.1007/BF02682046](https://doi.org/10.1007/BF02682046).
64. Tuttle JA, Chrismas BCR, Gibson OR, Barrington JH, Hughes DC, Castle PC, Metcalfe AJ, Midgley AW, Pearce O, Kabir C, Rayanmakar F, Al-Ali S, Lewis MP, Taylor L. The Hsp72 and Hsp90α mRNA responses to hot downhill running are reduced following a prior bout of hot downhill running, and occur concurrently within leukocytes and the vastus lateralis. *Front Physiol* 8: 473, 2017. doi:[10.3389/fphys.2017.00473](https://doi.org/10.3389/fphys.2017.00473).
65. Walsh NP, Gleeson M, Shephard RJ, Gleeson M, Woods JA, Bishop NC, Fleshner M, Green C, Pedersen BK, Hoffman-Goetz L, Rogers CJ, Northoff H, Abbasi A, Simon P. Position statement. Part one: immune function and exercise. *Exerc Immunol Rev* 17: 6–63, 2011.
66. Welc SS, Phillips NA, Oca-Cossio J, Wallet SM, Chen DL, Clanton TL. Hyperthermia increases interleukin-6 in mouse skeletal muscle. *Am J Physiol Cell Physiol* 303: C455–C466, 2012. doi:[10.1152/ajpcell.00028.2012](https://doi.org/10.1152/ajpcell.00028.2012).
67. Whitham M, Parker BL, Friedrichsen M, Hingst JR, Hjorth M, Hughes WE, Egan CL, Cron L, Watt KI, Kuchel RP, Jayasooriah N, Estevez E, Petzold T, Suter CM, Gregorevic P, Kiens B, Richter EA, James DE, Wojtaszewski JF, Febbraio MA. Extracellular vesicles provide a means for tissue crosstalk during exercise. *Cell Metab* 27: 237–251.e4, 2018. doi:[10.1016/j.cmet.2017.12.001](https://doi.org/10.1016/j.cmet.2017.12.001).
68. Williams DM, Dunsiger S, Ciccolo JT, Lewis BA, Albrecht AE, Marcus BH. Acute affective response to a moderate-intensity exercise stimulus predicts physical activity participation 6 and 12 months later. *Psychol Sport Exerc* 9: 231–245, 2008. doi:[10.1016/j.psychsport.2007.04.002](https://doi.org/10.1016/j.psychsport.2007.04.002).
69. Wong KL, Yeap WH, Tai JJ, Ong SM, Dang TM, Wong SC. The three human monocyte subsets: implications for health and disease. *Immunol Res* 53: 41–57, 2012. doi:[10.1007/s12026-012-8297-3](https://doi.org/10.1007/s12026-012-8297-3).
70. Yoshida M, Xia Y. Heat shock protein 90 as an endogenous protein enhancer of inducible nitric-oxide synthase. *J Biol Chem* 278: 36953–36958, 2003. doi:[10.1074/jbc.M305214200](https://doi.org/10.1074/jbc.M305214200).
71. Zawada AM, Fell LH, Untersteller K, Seiler S, Rogacev KS, Fliser D, Ziegler-Heitbrock L, Heine GH. Comparison of two different strategies for human monocyte subsets gating within the large-scale prospective CARE FOR HOME Study. *Cytometry A* 87: 750–758, 2015. doi:[10.1002/cyto.a.22703](https://doi.org/10.1002/cyto.a.22703).